flash, but any other tampering with the box to figure out how it works—for example, if we try to examine the sphere's color composition or chemical makeup before the door is opened—will also cause it to flash. In other words, we can't analyze the supposed random selection of red or blue because any such attempt will contaminate the very experiment we are trying to carry out. It's as if I told you I'm really a blonde, but I become a redhead whenever you or anyone or anything looks at my hair or analyzes it in any way. How could you ever prove me wrong? Your tiny green men are pretty clever—they've set things up so their ruse can't be unmasked. Now, go and play with your little boxes while I enjoy a little peace and quiet."

It would seem that Scully has this one soundly wrapped up on the side of science. Yet, here's the thing. Quantum mechanicians—scientists, not aliens—have for nearly eighty years been making claims about how the universe works that closely parallel those described in the letter. And the rub is that there is now strong scientific evidence that a viewpoint along the lines of Mulder's—not Scully's—is supported by the data. For instance, according to quantum mechanics, a particle can hang in a state of limbo between having one or another particular property—like an "alien" sphere hovering between flashing red and flashing blue before the door to its box is opened—and only when the particle is looked at (measured) does it randomly commit to one definite property or another. As if this weren't strange enough, quantum mechanics also predicts that there can be connections between particles, similar to those claimed to exist between the alien spheres. Two particles can be so entwined by quantum effects that their random selection of one property or another is correlated: just as each of the alien spheres chooses randomly between red and blue and yet, somehow, the colors chosen by spheres in boxes with the same number are correlated (both flashing red or both flashing blue), the properties chosen randomly by two particles, even if they are far apart in space, can similarly be aligned perfectly. Roughly speaking, even though the two particles are widely separated, quantum mechanics shows that whatever one particle does, the other will do too.

As a concrete example, if you are wearing a pair of sunglasses, quantum mechanics shows that there is a 50–50 chance that a particular photon—like one that is reflected toward you from the surface of a lake or from an asphalt roadway—will make it through your glare-reducing